EXAM PAPERS PRACTICE

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Detailed mark scheme

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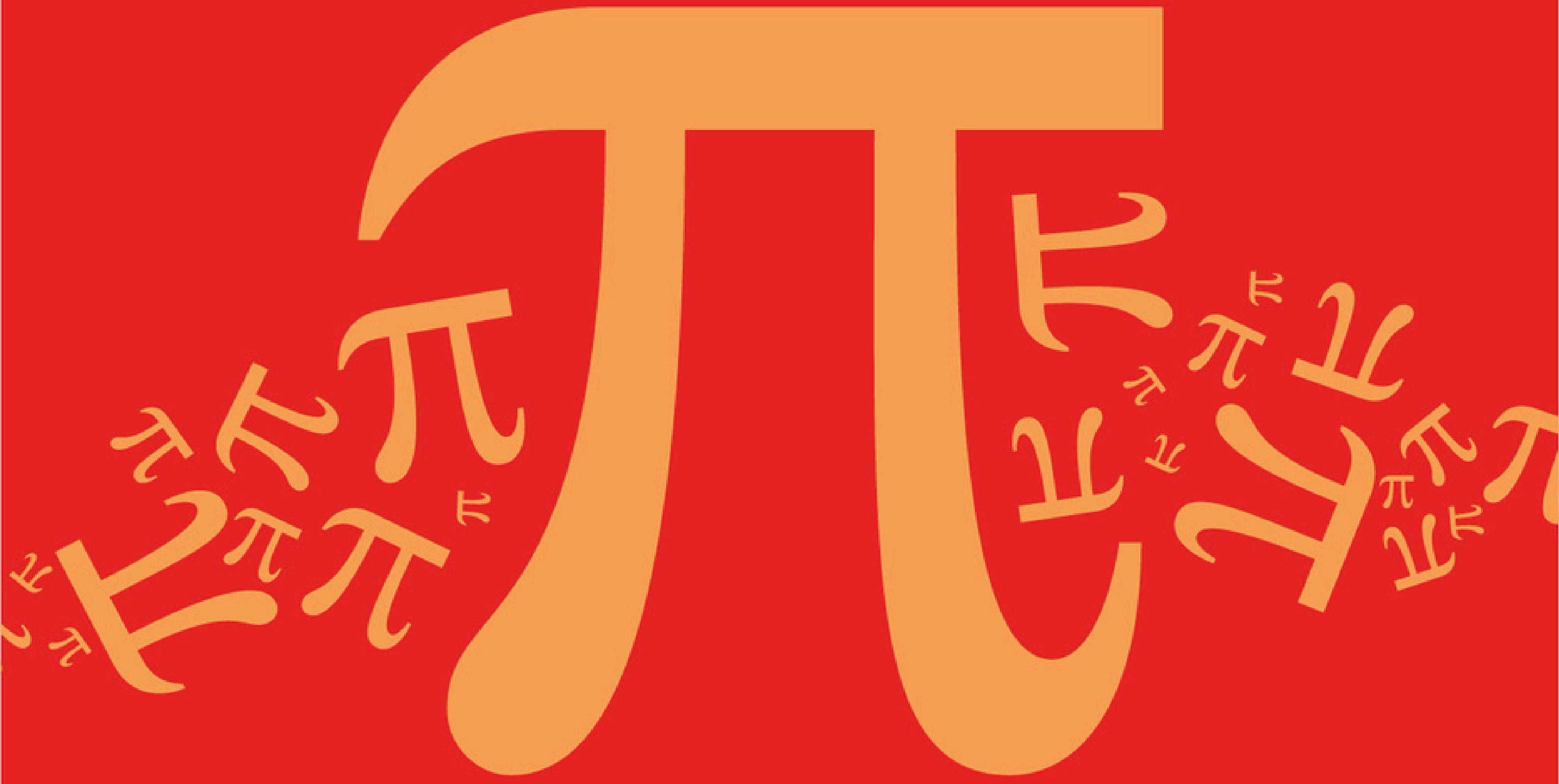
AQA A Level Further Mathematics (7367)



Topic **DA: Graphs**

Sub topic **DA1: Terminology for Graphs**

Topic Questions

EXAM PAPERS PRACTICE

AQA A Level Further Mathematics (7367)

Mark Scheme

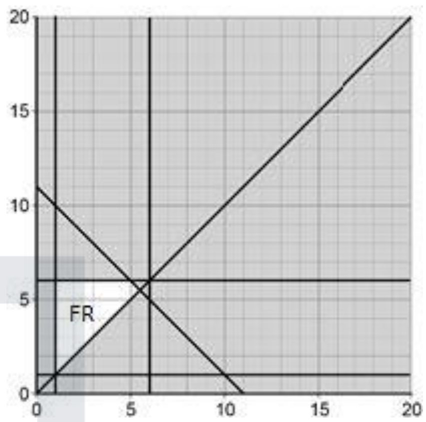
Topic	**DA: Graphs**
Sub topic	**DA2: Graph Properties (Eulerian/Hamiltonian)**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

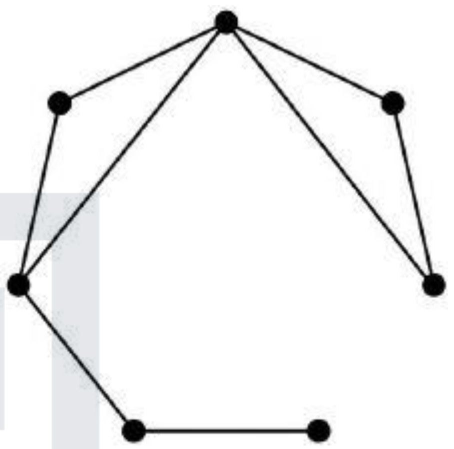
Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)(i)	Draws $y = x$	AO1.1a	M1	
	Clearly labels (or shows by shading) the correct feasible region	AO1.1b	A1	
(ii)	Calculates the correct value of $5x + 4y$ at (5, 6) or (5.5, 5.5)	AO1.1a	M1	Value at (5, 6) = 49 Value at (5.5, 5.5) = 49.5
	Finds the correct maximum value of $5x + 4y$	AO1.1b	A1	Maximum value is 49.5
(b)(i)	Explains the implication of the graph being connected Must refer explicitly to ‘connected’	AO1.2	B1	The graph is connected, so no vertex can have degree 0.
(ii)	Explains the implication of the graph being simple Must refer explicitly to ‘simple’	AO1.2	B1	The graph is simple, so each vertex can only connect at most once to each of the 6 other vertices.
(iii)	Explains why x and y cannot both be 6	AO2.4	E1	If x and y are both 6 then there would be two vertices that are both adjacent to the other five vertices.
	Completes rigorous argument for $x + y \leq 11$	AO2.1	R1	However, this would mean that there would be no vertex of degree 1 so one of x or y has to be at most 5. Hence $x + y \leq 5 + 6$
(iv)	Identifies the need for integer solutions	AO3.5b	B1	x and y must both be integers

	Marking Instructions	AO	Marks	Typical Solution
(c)(i)	Uses degree sum = 2 × 8 to deduce $x + y$	AO2.2a	M1	$1 + 2 + 3 + v + w + x + y = 2 \times 8$ $16 - (1 + 2 + 3 + 4) = x + y$ $x + y = 6$

	Infers there are multiple solutions and finds two correct feasible pairs of x and y values	AO2.2b	M1	$x = 1, y = 5$; $x = 2, y = 4$; $x = 3, y = 3$.
	Finds all three pairs and no others Condone pairs written as coordinates	AO1.1b	A1	
(ii)	Starts to show configuration of G by drawing a connected graph with exactly 8 edges OR by listing/labelling the vertex degrees: 1, 2, 2, 2, 2, 3, 4	AO3.1	M1	
	Draws a correct graph with degrees: 1, 2, 2, 2, 2, 3, 4 NB Multiple correct solutions, check by counting degrees	AO3.2a	A1	
				Total 14 marks

Q2.

(a)

$BD+FH$ = 210+210 = 420
$BF+DH$ = 200+180 = 380
$BH+DF$ = 260+340 = 600

These 3 sets of pairs

M1

3 correct totals, 2 correct totals

A2,1

(MIN) = 2430 + 380

2430 + their smallest of three pair totals

m1

= 2810

CSO

A1

5

(b) 2430 + 340 (DF) = 2770

2430 + their DF

B1F

1

(c) (i) 2430 + 180 (DH) = 2610

2430 + their min (must have scored M1)

B1F

1

(ii) B, F only

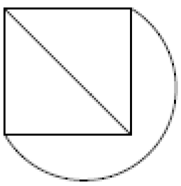
B1

1

[8]

Q3.

(a)



E1

1

(b) (i) 28

B1

(ii) Odd number of edges at (all) vertices

Must see the word odd, not just 7

E1

2

(c) (i) $\frac{n(n-1)}{2}$ OE

B1

(ii) $n - 1$

B1

(iii) n must be odd

Must have n in their answer

E1

(iv) $n = 3$

Must have n in their answer

B1

4

[7]

Q4.

(a) A vertex / vertices of odd order (A, B, G, H) OE

Condone statement of non-Eulerian graph

E1

1

(b)

$$\left.\begin{array}{l} AB + GH = (180 + 165) = 345 \\ AG + BH = (90 + 210) = 300 \\ AH + BG = (150 + 210) = 360 \end{array}\right\}$$

These 3 correct sets of pairs

M1

3 correct totals, 2 correct totals

A2,1

Dist 1215 + 300 PI

1215 + their smallest

M1

= 1515

CSO

A1

5

(c) (i) 3

B1

1

(ii) 2

B1

1

[8]

Q5.

(a) (i) 10

B1

1

(ii) 4

B1

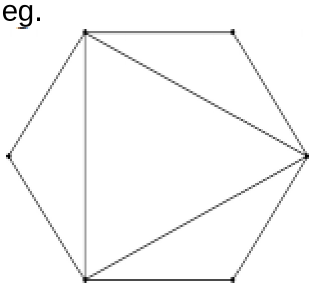
1

(iii) 5

B1

1

(b) eg.



Simple *graph, 6 vertices*

M1

Eulerian graph with 9 edges

A1

2

[5]

Q6.

(a) Max 5

B1

Min 1

Do not allow 1° or 5°

B1

2

(b) $4x - 12 \geq 1$ (or > 0)

$\left(x \leq \frac{13}{4}\right)$

Or

$4x - 12 \leq 5$ (or < 6)

$\left(x \leq \frac{17}{4}\right)$

Or

$2x - 4 \leq 5$ (or < 6)

$x \leq \frac{9}{2}$

Any one of these inequalities

OR

Exhaustive check of all values from 1 to 5 inclusive, condone one omission.

M1

$x = 4$

First inequality and one of the other two,
or completely correct exhaustive check,
and $x = 4$

A1

2

Alternative solution

Sum of degrees = $11x - 24$ must be even

$\Rightarrow$ x is even

$x - 2 > 0$ $\Rightarrow$ $x > 2$

M1

$x \leq 5$

Hence $x = 4$

A1

[4]

Q7.

(a) Min MST

4 edges

M1

= 8 + 10 +10 + 11 = 39

A1

2

(b) Max MST = 8 + 17 +17 +18

8 + 18 + 2 others

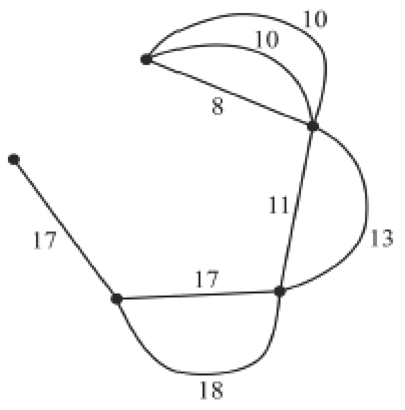
M1

= 60

A1

2

(c)



Connected graph with 5 vertices (all edges numbered, from G)

M1

MST = 53
8, 11, 17, 17 or 8, 10, 17, 18

A1

other edges OE

A1

(other possibilities not shown)
(all edges numbered, from G)

3

[7]

Q8.

(a) eg $A \quad B \quad C \quad D \quad E \quad F \quad A$

Any tour ABA or better, any start vertex but not revisiting a vertex
May be shown in a labelled diagram of a cycle (eg triangle ABC)

M1

With all vertices visited
May be shown in a labelled diagram of a cycle

A1

2

(b) (i) $F \quad D \quad C \quad A \quad B \quad E \quad F$
(20) (15) (5) (25) (15) (15)
(= 95) **AG**

Any tour, start/finish at F

M1

Visits all vertices

m1

Correct order

A1

3

If solution shown solely on matrix, then order of selection of vertices must be shown

(ii) Tour

"It's an answer", "a cycle", "it works"
"it's possible ..."

E1

May be improved on

"Can't be worse", "not necessarily best",
"could be improved"
Not "can be improved"

E1

2

(c) $F \quad E \quad C \quad A \quad B \quad D \quad F$

Tour $FE(ABCD$ in any order with B before $D)F$

M1

(30) (7) (5) (25) (11) (10)

Correct order

A1

= 88

B1

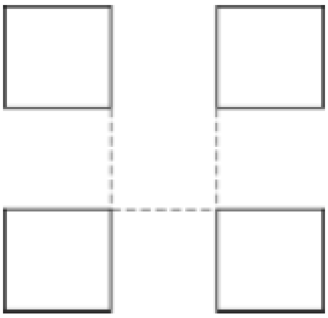
3

If solution shown solely on matrix, order of selection of vertices must be shown

[10]

Q9.

(a) (i)

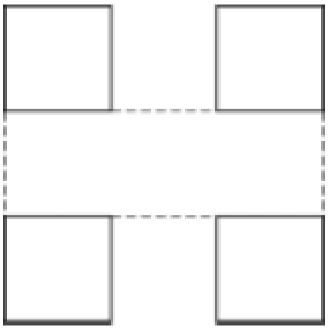


OE

B1

1

(ii)



4 edges

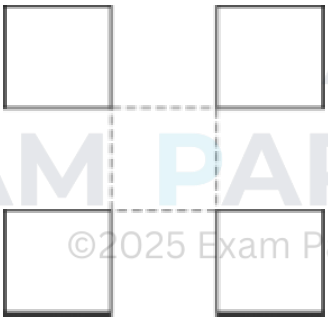
M1

OE
***Note**: new edges must meet each square at vertices on the opposite ends of a side of the square*
eg

A1

2

(iii)



4 edges

M1

Eulerian (all vertices are of even order)

A1

2

(b) (i) n odd

($n \pm 1$) even

B1

1

(ii) (Triangle) $n = 3$

Triangle, stated or drawn, scores B1

B2

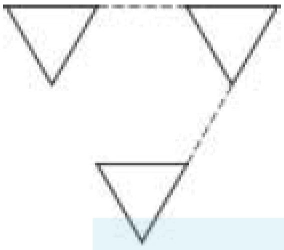
2

[8]

Q10.

(a) (i) 2

B1



OE

B1

2

(ii) 3

B1

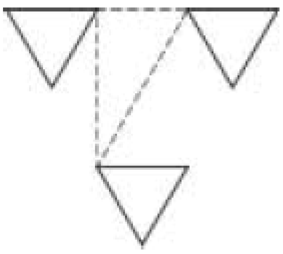
OE

B1

2

(iii) 3

B1

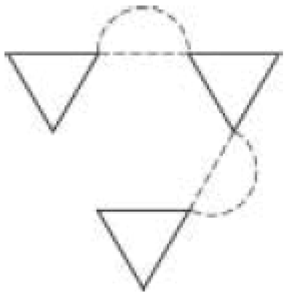


OE

B1

2

SC
4



OE

B1 (must have number and diagram)

(b) (i) n is odd

B1

1

(ii) 3 (only)

B1

1

[8]

Q11.

(a) (i) 9

B1

1

(ii) $n - 1$

B1

1

(b) (i) GI 5

9 edges

B1

AB 6

SCA

M1

EI 7

start with GI

A1

BD 8

~~EG 9~~

IJ 10

IJ fifth

A1

HJ 11

~~HI 12~~

111 1Z,

AF 13

DE 14

CG 15

all correct

A1

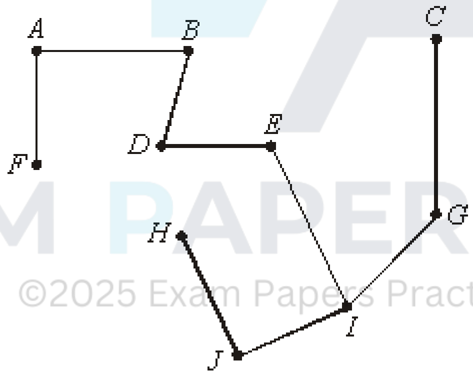
5

(ii) 89

B1

1

(iii)



9 edges

M1

A1

2

[10]

Q12.

(a) Odd vertices (at 4 B, C, I)

E1

1

(b) $AB + CI$ = 100 + 440 = 540

M1

$AC + BI$ = 150 + 450 = 600

A2,1,0

$AI + BC$ = 380 + 120 = 500

Repeat $AI + BC$

May be implied

E1

Distance 2090 + 500 = 2590

B1

5

(c) Route with

(3A), 2B, 2C, 3D, 2E, 2F, 3G, 1H, 2I, 1J

(16 → 21)

M1

= 18

A1

2

[8]

Q13.

(a) (i) $m - 1$

B1

1

(ii) $n \geq m - 1$

B1 for > or($n \geq m$) OE

B2

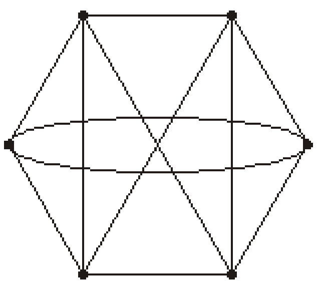
2

(b) $m(=n)$

B1

1

(c)



m = 6 and eulerian

M1

All correct

A1

2

[6]

Q14.

(a) Odd vertices (*ADFI*)

E1

1

(b) $AD + FI = 14 + 14 = 28$

M1

$AF + DI = 14 + 13 = 27$

$AI + DF = 11 + 17 = 28$

A2,1,0

$\therefore$ Repeat $AF + DI$

may be implied

E1

Distance = 87 + 27 = 114

B1

Route with

$3A, 1B, 2C, 2D, 3E, 2F, 1G, 1H, 2I$

17 vertices

B1

6

[7]

Q15.

(a) (i) 21

B1

1

(ii) 6

B1

1

(iii) 7

B1

1

(b) (i) All vertices are even

OE

E1

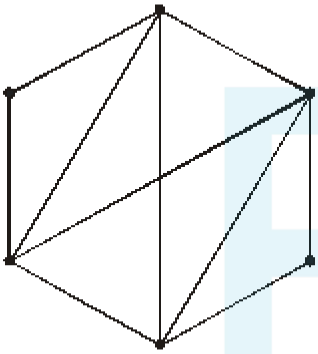
1

(ii) *n* odd

E1

1

(c)



Graph with 6 vertices

M1

A1

2

[7]

Q16.

(a) Min = 4 + 7 + 7 + 7

= 25

B1

1

(b) Min(H) = 4 + 7 + 7 + 7 + 8

= 33

B1

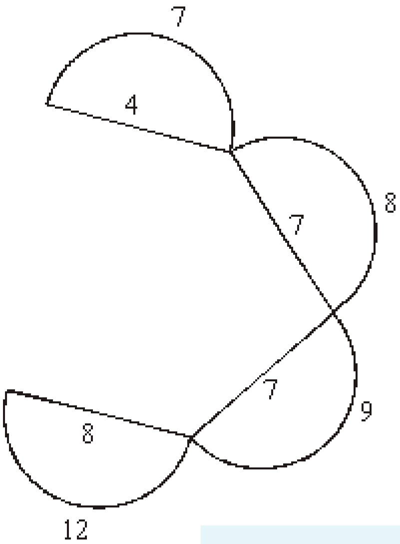
1

(c) Min(E) = Σ = 62

B1

1

(d)

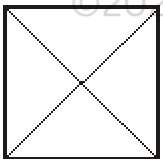


5 vertices	M1	
8 edges	m1	
All correct	A1	3

[6]

Q17.

(a)



Correct graph	M1	
4 vertices	A1	2

(b) (i) 15 B1 **1**

(ii) 5 B1 **1**

(iii) No, order of vertices is odd

B1 1

(c) (i) $\frac{n(n-1)}{2}$
OE

B1 1

(ii) $n - 1$

B1 1

(iii) n ODD
OE

B1 1 **[8]**

Q18.

(a) (i) odd vertices

E1 1

(ii) C, D, E, F
$CD + EF = 200 + 150 = 350$

M1

$CE + DF = 200 + 200 = 400$

A2,1,0

$CF + DE = 325 + 50 = 375$
Repeat CGD& EF

B1

Tour with

M1

$A2, B2, C2, D2, E2, F2, G3$

A1

Distance = 1500+ 350
= 1850

B1F 7

(b) $1 \times 3 \times 5 = 15$

M1A1

2

[10]

Q19.

(a) (i) 4

B1

1

(ii) 2

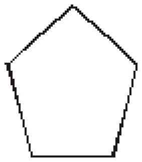
B1

1

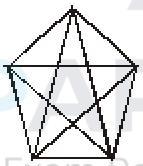
(iii) 5

B1

1

(b) 

or equivalent



or

B1

1

[4]

Q20.

(a) (i) $(ABDK) = 3$

B1

1

(ii) $MST = (n - 1)$

M1

= 15

A1

2

(b) (i) Odd vertices A and P

Route length = all edges + repeated edges (answer > 30)

M1 M1

Route length = all + (3)

A1

(33 + 3) = 36

B1

4

(ii) Repeated: AE, EL, LP

B1

1

(iii) E and L now odd

M1

Their(33) – 2 + 1 = 32

A1F

2

[10]

Q21.

(a) Graph 1: no – odd vertices (semi)

for considering vertices

M1B1

Graph 2: no – odd

B1

Graph 3: yes – even

B1

4

(b) Graph 1: 8 edges

for considering repeats on 1 or 2

allow listings of CP

M1/B1

Graph 2: $\begin{array}{l} 10 \text{ edges} \\ + \dfrac{2}{12} \text{repeats}(BC \text{ and } FE) \end{array}$

B1

Graph 3: $\begin{array}{l} 12 \text{ edges} \\ + \dfrac{0}{\underline{12}} \text{ repeats} \end{array}$

B1

4

[8]

Q22.

(a) (i) Odd vertices ($\Rightarrow$ Repeats)

Eularian not enough

E1

1

(ii) Odd *BCDE*

considering odds

M1

BC + *DE* = 14
BD + *CE* = 12.5 + 10.5 = 23
<u>*BE*</u> + <u>*CD*</u> = 9 + 4.5 = <u>13.5</u>

A2,1,0

Distance = 13.5 + 82
= 95.5

82 + their (mir)

M1

A1F

Route: *A B C D E F B E B D C A*

or equivalent

B1

6

(b) (i) 5 × 3 = 15

B1

1

(ii) (n – 1)(n – 3)(n – 5)...× 1

M1

(n – 1)(n – 3)

A1

2

[10]

Q23.

(a) 3

B1

1

(b) (i) $n - 1$

B1

1

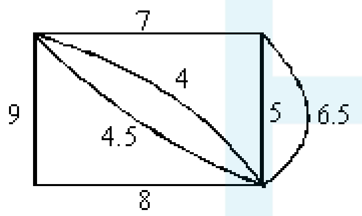
(ii) $(n - 1)!$

M1 for factorial

M1A1

2

(c)



M1

connected graph

M1

m.s.t. = 17

A1

all other arcs used and correct

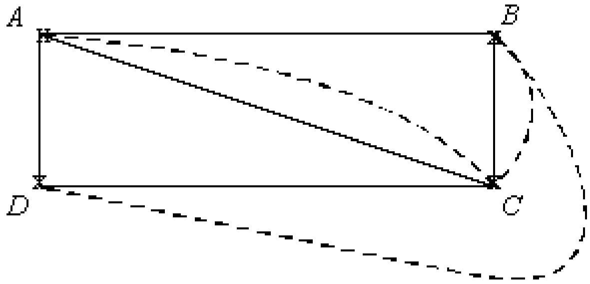
A1

3

[7]

Q24.

(a) (i)



correct graph

M1

All correct (may be directed)

A1 2

(ii) Odd vertices

B1 1

(b) $\frac{\text{Sum}}{2} =$ number of edges

B1 1

(c) Directed $A \rightarrow B \neq B \rightarrow A$

Or equivalent

E1 1

[5]

Q25.

(a) (i) Need 4 + 3 + 2 +1 edges

E1 1

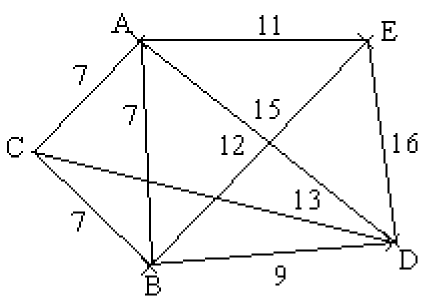
(b) (i) M.S.T. = 7 + 7 + 7 + 9 = 30

B1 1

(ii) Might include loop

E1 1

(iii)



Must include loop with

M1

7's

M1

No loop with 9 & 11

A1

3

[6]

Q26.

(a) Trace each edge once and only once, ending at starting point

E1

1

(b) (i) Odd vertices

E1

3

(ii) A and E

B1

(iii) ADBCDEBAE

or equiv.

B1

[4]

Q27.

(a) Closed path

E1

Every vertex once

E1

2

(b) 3! 6

M1A1

2

(c) (n – 1)!

M1 for (n – 1) or factorial

M1A1

2

[6]

Q28.

(a) (i) C - D - E - F

any route between C and F

M1

C - D - E - F offered as the shortest route

A1 2

(ii) eg *ABCDEFEDCEBFA*

Route starting and finishing at A, with C to F repeated

M1

Complete correct route

ft route C to F in (i)

A1F 2

(iii) Network redrawn with two directed edges BE

Edges BE must be directed, one each way

B1

eg *ABEBCDECBFEFA*

A route, starting and finishing at A, with BE and EB both included

M1

Complete correct route, this time with BC and EF repeated

A1 3

(b) The start and finish need not be the same

E1 for any two (or equivalent) comments

E1

There are exactly two odd vertices, C and F.

E1 for the third (or equivalent) comment

E1

Start at C (or F) and finish at F (or C).

Alternative

give the complete route eg FABFEBCEDC

(E2) 2

[9]

Q29.

(a) 1

B1 1

(b) 2

B1

e.g. *AB* and *ED*

B1 2

(c) (i) e.g. add *AB*

B1

Eulerian trail *AEFABCDB*

M1 A1 3

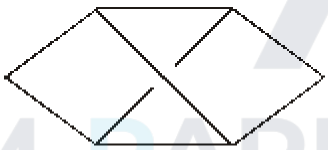
(ii) e.g. adding *AB* twice creates a connected graph with all degrees even

B1 1

(iii) 4

B1

e.g.



M1A1 3

[10]

Q30.

(a) E and H

Deduct 1 for error or omission

B2,1,0 2

(b) F, G and H

Deduct 1 for error or omission

B2,1,0 2

(c) E, F and G

Deduct 1 for error or omission

B2,1,0 2

(d) F and G

(or M1 A1)

2 cao, or sensible working towards wrong answer will earn ½

B2

2

[8]

Q31.

(a) (i) Sum of degrees = $5d + 7$

B1

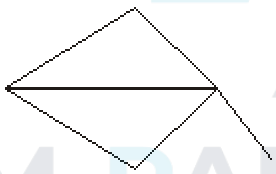
This must be even and so d must be odd

B1

2

(ii) $d \geq 3$ makes $d + 3$ too big, so $d = 1$:
e.g.

B1



M1A1

3

(b) (i) each vertex is joined once to at most the other 9 vertices

B1

1

(ii) If all the degrees were different they would be 0 1 2 3 … 9.

M1

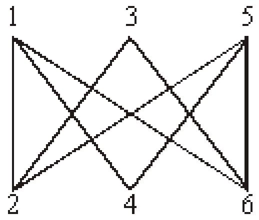
But that would give a vertex joined to none and another joined to them all – clearly impossible.

A1

2

[8]

Q32.

(a) 

M1 A1 2

(b) (i) e.g. 1234561

M1 A1 2

(ii) For even *n* ((and > 2))

B1 for "4 and 6"

(the ">2" not expected)

B2,1 2

(c) ½ *n*

B1 1

(d) <u>Case *n* even</u>:

M1

By (i) we need ½ n even;

i.e. *n* must be visible by 4

A1

<u>Case *n* odd</u>:

Some vertices always have odd degree, so not Eulerian

B1 3

[10]

Q33.

(a) *CADEBFMLKJC*

M1A1 2

(b) Neither

B1

More than two odd vertices

B1

2

(c) (i) *KL*

B1

1

(ii) Starting at *A*, say, we can only get to *B*

M1

along *DE*, but then we cannot get back.

(any sensible focus on the 'isthmus' DE)

A1

2

[7]

Q34.

(i) Maximum degree = 5, so $d \leq 1$

B1

1

(ii) Semi-Eulerian for $d = 1$

B1

This gives 2 odd degrees – in other case there are > 2 odd vertices.

(or d = 0 gives an isolated vertex)

B1

2

(iii) $d = 0$ gives an isolated vertex,

B1

d = 1 gives a 'dead-end'

B1

2

[5]

Q35.

ABCDHGFEA

M1

ABCGFEHDA

A1

A1 3 [3]

Q36.

(a) None is Eulerian as each has a vertex of odd degree.

B1 1

(b) Only G_2 is semi-Eulerian:

B1

e.g. LMIHLJHKMJI

M1 A1 3 [4]

Q37.

(a) Odd vertices *P Q R U*.

B1

Pairings	**Minimum extra**
PQ RU	2, 1 + 2 $\Rightarrow$ 5

M1

PR QU	2 + 2, 2 + 2 $\Rightarrow$ 8

A1

PU QR	3 + 2, 2 $\Rightarrow$ 7

A1

Therefore minimum closed Eularian walk is obtained by repeating *PQ* and *RSU* to give a total of 34 + 5 = 39

A1 5

(b) *S R Q P T V U S*

M1 A1 A1

1 + 2 + 2 + 4 + 2 + 3 + 2 = 16

B1 4

(c) (i) Two edges at *U* must be used.

B1

(ii) In the reduced network the 'bottleneck' at *S* prevents a Hamiltonian cycle.

B1

(iii) (i) & (ii) imply that the length of a Hamiltonian cycle

M1

$\geq 4 + 3 + 1 + 2 + 2 + 2 + 2 = 16$

A1

4

(d) (i) Hamiltonian route clearly required.

M1

Shortest time = 16 + 20 × 6 = 136 seconds

ft (and allow 156)

A1ft

2

(ii) From (a) the shortest route uses 16 edges and takes time = 39 + 20 × (no of times route passes through a centre)

M1

ft

A1ft

= 39 + 20 × 15 = 339 seconds

(allow 359)

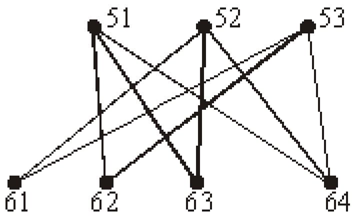
A1

3

[18]

Q38.

(a)



a correctly labelled graph in the style shown can earn the marks for (a) and (b)

M1 A1

2

(b) more than two odd vertices

B1

1

(c) e.g. 51 62 53 61

B1

none of length 4

B1

every edge is from {51, 52, 53} to {61, 62, 63, 64}, so to get back where you

M1

started takes an even number.

or 'bipartite'

A1

4

(d) A Hamiltonian cycle would be a closed path with seven edges.

or alternative

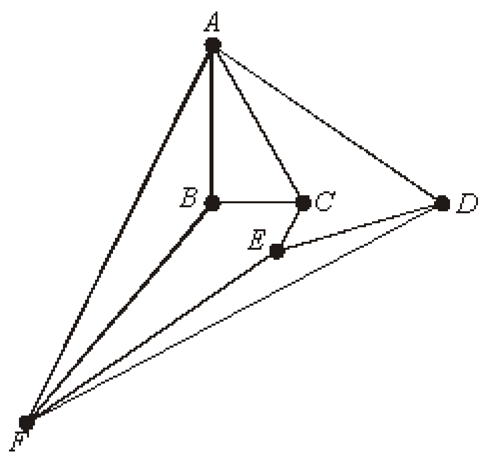
M1 A1

2

[9]

Q39.

(a)



M1 A1 A1

3

(b) Neither

B1ft

2

More than 2 vertices of odd degree

B1ft

(c) (e.g.) *ACEDBFA*

M1ft A1ft

2

(d) Hamiltonian, to enable a round tour of all

M1

the rooms without revisiting any

Or any sensible alternative

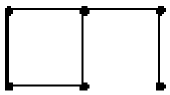
A1

2

[9]

Q40.

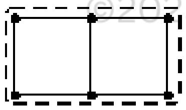
(a) eg



B1

1

(b) 2; eg

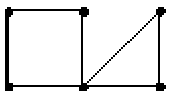


any Hamiltonian extension

B1 B1

2

(c) 2; eg



any Eulerian extension

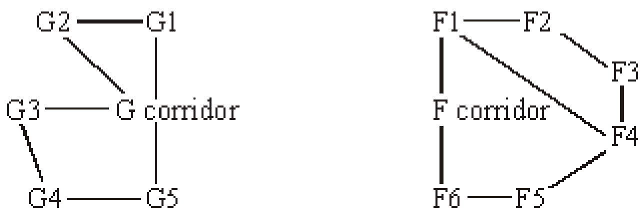
B1 B1

2

[5]

Q41.

(a) & (b)



Corridor vertex OK

M1

ground floor

A1

M1 F1 OK

M1

4

first floor

A1

(c) Ground floor is Eulerian since all vertices are even

B1

First floor is semi-Eulerian since there are two odd vertices, F1 and F4

B1

Can pass through all doors once only in a tour of the ground floor.
Can do the same on first floor only if starting/finishing at F1/4.
Otherwise the door connecting F1 and F4 must be used twice

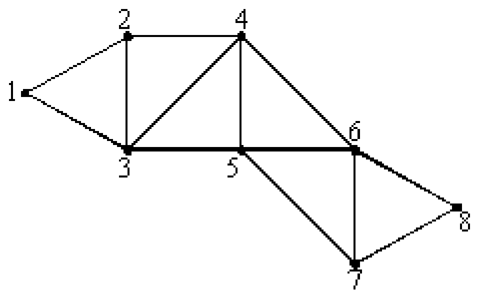
B1

3

[7]

Q42.

(a)



M1

two Δ

A1

Rest

A1

(b) Two odd vertices, 2 and 7

B1

e.g. 2312 4354657687

M1A1

(c) eg 124687531

M1A1

(d) 13578, 13568, 13468, 12468

M1

for two

A1

Rest

A1

[11]

Q43.

(a) Sum = 1lm = even ∴ m = even
Also 0 < 2m < 6 ∴ m =2

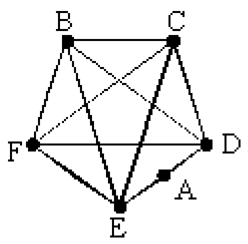
$\frac{1}{2}$ *for sensible incomplete argument*

B1

B1

2

(b) e.g.



M1

SA = 2 and attempt at rest

A1

Rest

A1

(c) 10 (choices for A’s pair of neighbours)

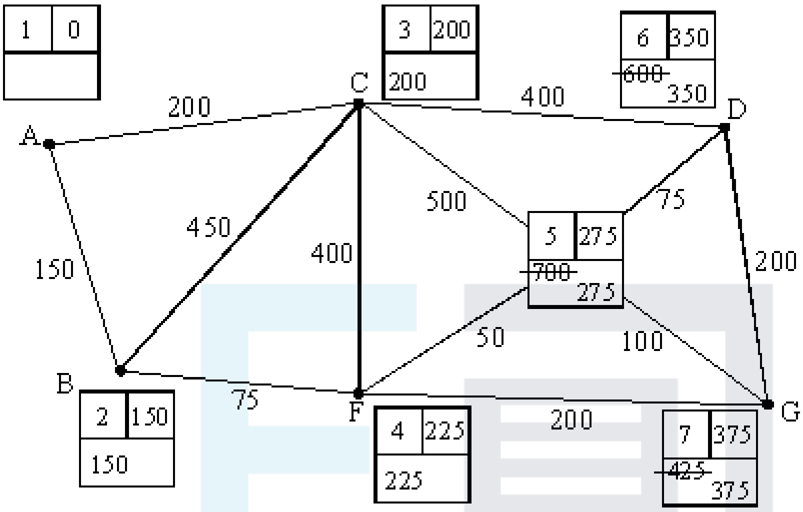
M1A1

5

[7]

Q44.

(a)



Dijkstra M1

working values A1

Labels A1

Shortest route (traceback): ABFEG, with length 375

B1 B1

5

(b) odd vertices

B1

BC/DG: 350 + 175 = 525

B1

BD/CG: 200, 550

B1

BG/CD: 225, 400

B1

e.g. ACABCDEDGEGFECFBA

B1

Length = 2800 + 525 = 3325

B1

6

[11]

Q45.

(a) Closed (back to start)

B1

Every vertex exactly once

B1 B1

(b) $(4 \times 3 \times 3 \times 2 \times 2)/2 = 72$

M1 sca

A1

(c) Can’t avoid edge crossing(s)

B1

$K_{3,3}$ is a subgraph. (eg. Delete D and S + edges)

B1

[7]

EXAM PAPERS PRACTICE

Examiner reports

Q1.

This was a question which allowed students to demonstrate their understanding and application of mathematics in an unfamiliar situation, which is a key feature of the new specification. The majority of students scored 2/2 in part (a)(i). The most common error was to select an incorrect region as the feasible region. Some students did not draw the line $y = x$. Less than half of the students scored 2/2 in (a)(ii), with the most common mistakes being selecting the wrong vertex. The best solutions included an objective line which made the selection of the optimal vertex easier.

Nearly three quarters of students were able to explain the term ‘connected’ and relate it to the inequalities in (b)(i). However, less than half of students were able to correctly explain how the graph being simple related to the inequalities. Errors that were made included no direct reference to the degree of the vertices or the particular property of the graph that was being used.

Only 6% on students scored both marks in (b)(iii), with many students erroneously restating what they had written in (b)(ii). There were a significant number of no responses on this part. 10% of students scored the mark in part (b)(iv), with common errors being the adding together of two inequalities or the stating of a previous inequality.

Only a third of students scored any marks in (c)(i), with the vast majority not using the sum of the degrees being equal to twice the number of edges in the graph. Many answers did not write the values of x and y as pairs, and some solutions included pairs of x and y values that did not satisfy all of the inequalities.

More than half of students were able to score at least one mark in (c)(ii), with many students scoring 2/2 having not scored full marks on (c)(i). The better solutions included writing down the degree of each vertex next to the respective vertex. The loss of marks was most commonly due to having too few or too many edges on the graph.

Q2.

Part (a) was well-answered – students had been well-drilled in the rote use of an elementary algorithm, although arithmetical accuracy was sometimes weak. However, in the remainder of the question, students were required to do a little elementary analysis and thinking and many failed completely. It appears that these modified situations are not handled well. Students need to spend time studying semi-Eulerian situations.

Q3.

This questioned differentiated well between the students. Many students showed a lack of understanding or knowledge of graph theory and were unable to answer this question correctly and weaker students found the abstractness of this question a too much of a challenge.

The most common problem with par (b)(ii) was an incomplete or inaccurate description of vertices of an odd order, with students often talking about “odd vertices” or an “odd number of vertices”.

Marks were quite often lost in part (c) because of a failure to note the implications of the phrase “state in terms of n” and part (c)(iv) identified the strongest students.

Q4.

Parts (a) and (b) were answered well. The only significant error was to calculate the value of BG incorrectly - the least length being 210, not 225.

Part (c) did cause problems, usually part (c)(i) where 2 was the common wrong answer.

Q5.

Very few candidates failed to score marks but very few scored full marks. The first mark was often dropped in part (a). Few candidates got both marks in part (b) - many joined two vertices with more than one edge; a majority of those who surmounted this hurdle then presented vertices of odd order.

Q6.

Very few marks were scored. Around half the candidates appeared to think this question was about hexagons and certainly about the sizes of angles in degrees. From those who correctly interpreted the question the answers 2 and 6 to part (a) were about as common as the correct answers 1 and 5.

Most scoring attempts at part (b) were by numerical checking rather than the use of algebraic inequalities and revealed that most candidates had little or no concept of ‘show’, let alone ‘proof’ in this question.

Q7.

The usual mark scored was 4 out of 7, with very few candidates getting part (b) or the final mark in part (c) due to adding in extra edges and changing the MST.

Part (a) was usually correct.

Part (b) was usually incorrect, with 13 + 17 + 17 + 18 = 65 being the almost universal answer. The very few who answered correctly were by no means always drawn from those scoring highest on the rest of the paper.

In part (c), most of the candidature spotted one of the two correct MSTs. A number of candidates elected to draw only a correct MST. Those candidates who attempted to complete the diagram frequently forfeited marks as their extra edges impacted on their MST. A few candidates drew separate diagrams.

Q8.

This work was well known, continuing the improvement from previous series. The weakest candidates often failed to complete tours. There were very few answers presented in matrix form – a considerable proportion of these failed to provide indication of order. In part (a), a surprising number of candidates omitted to give an example, opting to give a general description or definition of a tour. Part (b)(i) was usually correct. In part (b)(ii), as with similar past questions, there was a majority view that the current figure *can* be improved rather than *might* be. Some still failed to pick up the ‘tour’ mark. There were many correct responses to part (c) of the question.

Q9.

This question discriminated well. Despite the comparative ease of marking the short, sharp answers, it didn’t appear that weak candidates could ‘guess’ their way to a reasonable mark. In part (a), full marks were very rare – the preserve of the generally most able candidates. Correct answers to parts (a)(ii) and (a)(iii) were often interchanged. Correct answers were also very rare in part (b). Many knew the node property required in part (b)(i) but failed to convert this into an acceptable statement about n. '4' was a more popular answer to part (b)(ii) than the correct answer. Average and weak candidates often provided answers consisting of ever more complicated algebraic expressions.

Q10.

Although this question discriminated between candidates, it was answered poorly by many. Candidates must be able to apply their knowledge in questions, but graph theory as a topic is important. This part of the specification appears to have been overlooked by many centres. The understanding of Hamiltonian and Eulerian was often confused.

Q11.

This question was a good source of marks for virtually all candidates, with few scripts using the wrong algorithm. Candidates who listed all the edges in part (b) were successful, although a common mistake was including edge FH. Some diagrams of minimum spanning trees were not labelled at the vertices.

Q12.

The majority of candidates correctly answered part (a). In part (b) candidates realised that the method involved pairing odd vertices, however many candidates failed to find the minimum pairing connecting the odd vertices. Candidates must realise that they must consider the shortest distances connecting vertices when solving a Chinese postman problem. Many candidates who obtained the correct minimum pairing AI + BC failed to obtain the correct answer of 500.

Part (c) was poorly answered. Candidates should realise that the purpose of pairing odd vertices is to add extra edges to the network. This makes the network Eulerian and it is immediately apparent as to the number of times each vertex would appear in an optimum route.

Q13.

This question discriminated between candidates, with many showing little understanding of graph theory. Part (a)(ii) was beyond nearly all candidates and in the final part a significant number of candidates stated that all vertices needed to be of even order and then drew a graph with odd vertices. Candidates should expect to be tested on all different aspects of graph theory in the specification.

Q14.

Nearly all candidates were able to answer part (a) correctly. In part (b), candidates must realise that for a Chinese postman problem involving 4 odd vertices, they have to show the 3 pairings **and** justify that they have selected the optimum pairing. A number of candidates thought that the sum of the 3 pairings was the same and that any pairing could be chosen. Some candidates who correctly chose the correct pairing to repeat failed to find a corresponding route. Candidates are encouraged to list the orders of the vertices after the repeats have been included. They would then be able to check if they had a correct route corresponding to their minimum distance.

Q15.

This question discriminated between candidates. Although candidates must be able to apply their knowledge in questions, graph theory as a topic is important. The majority of candidates were able to attempt all parts but parts (a)(i) and (b) (ii) proved to be the most challenging.

Q16.

This question proved to be a good discriminator, requiring candidates to think of an abstract network. Part (a) was well answered, but correct answers to parts (b) and (c) were rarely seen. Candidates started part (d) well by drawing a network with 5 vertices and a minimum spanning tree of 26 units, but often went astray when adding the extra arcs, ending up with a graph for which the minimum spanning tree had a length of 25 units.

Q17.

This question was the least well answered on the paper. Although candidates must be able to apply their knowledge in questions, this tested graph theory itself and not the application of graph theory. The majority of candidates were able to answer parts (a), (b)(ii) and (c)(ii). Complete networks were not understood. The total number of edges was poorly answered. Parts (b)(iii) and (c)(iii) tested the understanding of Eulerian graphs and whilst candidates realised that the order of vertices was important, many failed to score the mark in part (c)(iii).

Q18.

The majority of candidates correctly answered part (a)(i). In part (a)(ii) candidates realised that the method involved pairing odd vertices, however many candidates failed to find the minimum pairing connecting $CD+EF$ and $CF+DE$. Candidates should realise that they must consider the shortest paths connecting vertices when solving a Chinese postman problem. Many candidates who obtained the correct minimum of 350 by pairing $CD+EF$ failed to use this fact and repeated the direct connection of CD. Part (b) was poorly answered. It is a standard piece of bookwork that candidates should know.

Q19.

Candidates generally scored high marks on this question, although full marks were rarely seen. Many candidates gave the wrong answer of 72 to part (a)(ii).

Q20.

Part (a) was well answered.

Part (b) proved to be more challenging. Candidates were able to make a good attempt at part (i), although a significant number used 16 + 2 or 3. In part (ii), a common mistake was to omit the edge EL. Part (iii) proved to be a good discriminator. Candidates failed to realise that by removing AE and LP, then E and L would become the odd vertices.

Q21.

This question was well answered with many candidates scoring the full 8 marks. In the first graph in part (a) many candidates gave the answer of semi-Eulerian which was acceptable and in part (b) a number of candidates listed the Chinese postman route when all that was required was the length. These candidates were not penalised.

Q22.

Although candidates realised that an optimal 'Chinese postman' route depended on odd vertices, there was a disappointing number who failed to pair off the odd nodes correctly to arrive at the correct minimum repetition. A surprising number of candidates who did arrive at the correct numerical answer failed to write down a route. Part (b) proved to be the most difficult part of the paper with only a few candidates obtaining the correct answer for part (b)(ii).

Q23.

Most candidates made a reasonable attempt at this question but very few scored full marks. Many did not see the connection between parts (a) and (b)(i). In part (c) many candidates were able to draw a correct minimum spanning tree but very few were able to add on the remaining arcs successfully.

Q24.

The question was answered well but a significant number of candidates failed to answer part (c) correctly, with many believing that a look at a vertex was a correct answer.

Q25.

This question proved to be discriminating between the candidates. Although all candidates were able to attempt all of the question there were very few correct answers to part (b)(iii).

Q26.

The question was well answered but a significant number of candidates were too vague in their description of an Eulerian cycle to score the mark in part (a).

Q27.

Although this question was a standard piece of book work, correct solutions were rarely seen. In part (a), candidates needed to state that a Hamiltonian cycle is a closed path, visiting every vertex once. Part (b) only used four vertices so that candidates could, if they wanted, list all cycles. There were only two correct answers to part (c).

Q28.

The candidate was able to work with the odd vertices but didn't consider which was the shortest route to connect these two vertices. The same difficulty arose in answering part (a)(iii). Part (b) was correctly answered.

Q32.

Most candidates could draw $\mathbf{G}_6$, find a Hamiltonian cycle of it, and state that $\mathbf{G}_n$ is only Hamiltonian for even n (they were not penalised for failing to notice that $\mathbf{G}_2$ is not Hamiltonian). Most also recognised that $\mathbf{G}_6 = \mathbf{K}_{3,3}$ and found the correct degrees in part (c).

Q33.

The first parts of the question were well done, although many candidates failed to make their Hamiltonian cycles return to C (no marks were lost for this). Also many candidates failed to earn the final two marks. In general if a candidate is asked "why is this graph not

Hamiltonian" then the tautology "because it has no cycle visiting all the vertices" hardly answers the question because it makes no specific reference to the given graph. Here a simple answer involving the disconnecting edge DE was required.

Q35.

The more standard graph meant that the performances on the Hamiltonian cycles was great improvement on previous papers, with most candidates scoring well.

Q36.

The understanding of graph theory shown by candidates' answers to this question was a very pleasant surprise.

Q37.

Again this was very well done. Candidates displayed a good working knowledge of the Chinese postperson algorithm and the nearest neighbour algorithm, and in the final part they applied their results correctly.

Q38.

This question was quite well done.

Most candidates struggled to find a convincing reason why the graph was not Hamiltonian. The examiner had tried to be helpful: in part (b) we saw that all cycles had to be even, but a Hamiltonian cycle would have to use 7 edges!

Q40.

The answers here were generally good, but only a minority correctly spotted that two additional edges were sufficient to make the graph Eulerian.

Q42.

As usual some candidates confused Eulerian trails and Hamiltonian cycles. Even those who understood the terms often struggled to stipulate them.

Q43.

The reasoning given in part (a) was rather poor. All that was needed was "The total of $11m$ must be even, so m must be even. Hence for all degrees to be less than 6 we need 2 = m". Alternatively, many candidates correctly used the Hamiltonian cycle method to show non-planarity. As intended, the last part proved to discriminate well.

Q44.

In part (a) of this question evidence of the required algorithmic work was often lacking. Only a minority of candidates systematically considered all possible pairings of odd vertices in part (b), as required.

Q45.

Part (a) was well answered, but part (b) was not. Very few candidates were able to perform this simple count. Most candidates knew Kuratowski's theorem in part (c)